



Validation and application of analytical method for glyphosate and glufosinate in foods by liquid chromatography-tandem mass spectrometry

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ABSTRACT

A reliable and sensitive method was developed for simultaneous determination of glyphosate and glufosinate in various food products by liquid chromatography-tandem mass spectrometry. Based on extraction, derivatization with 9-fluorenylmethylchloroformate and purification on solid phase extraction column, quantification was done by using isotopic-labeled analytes as internal standard and calibration in matrix. Good selectivity and sensitivity were achieved with a limit of quantification of 5 µg/kg. The recoveries of these two pesticides ranged from 91% to 114% with inter-day and relative standard deviation of 3.8–6.1% in five matrices of cereal group spiked at 5, 10, and 20 µg/kg. An accuracy profile was performed for method validation, demonstrating the accuracy and precision of the method for the studied food groups. The verification results in expanded food groups indicated extensive applicability for the analysis of glyphosate and glufosinate. Finally, the developed method was applied to analyze 136 food samples including milk-based baby foods from the French Agency for Food, Environmental and Occupational Health & Safety. Glyphosate residues were detected in two breakfast cereal samples (6.0 and 34 µg/kg). Glufosinate residues were found in a sample of boiled potatoes (9.8 µg/kg). No residues were detected in the other samples, including milk-based baby foods with limits of detection ranging from 1 to 2 µg/kg. The method has been applied for routine national monitoring of glyphosate and glufosinate in various foods.

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1. Introduction

Glyphosate [N-(phosphonomethyl) glycine] and glufosinate [DL-homoalanine-4-yl(methyl) phosphonic acid] are non-selective, broad spectrum herbicides and widely used for weed control in agriculture, forestry, and urban applications [1,2]. They are considered to be non-persistent and safe in environment for their rapid degradation in water and soil. However, glyphosate and glufosinate have attracted a high level of public and scientific interest, because they are frequently detected in plants and soil, even in foods and human urine [3–7]. In order to better estimate human dietary exposure, it is necessary to expand

the analysis of these two pesticides in different kinds of food, including baby foods.

Glyphosate and glufosinate are belonging to the amino acid class of pesticides. These two compounds are challenging to analyze at low concentration level due to their amphotericity, highly polarity, non-volatility, and lack of chromophores or fluorophores [8]. Until now, many methods have been applied for the analysis of glyphosate and glufosinate, mainly including gas chromatography coupled with mass spectrometry (GC-MS) [9], or gas chromatography tandem mass spectrometry (GC-MS/MS) [10], liquid chromatography coupled with fluorescence (LC-FLD) [11], ultraviolet (LC-UV) [12] and mass spectrometry (LC-MS) [13], liquid chromatography-tandem mass spectrometry (LC-MS/MS) [14], and so on. To reduce method complexity and time consumption, the capillary electrophoresis and hydrophobic interaction chromatography, inductively coupled plasma MS (ICP-MS) were used to analyze glyphosate without prior derivatization achieving sepa-

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ration from matrix component [15–17]. In most of these methods, LC–MS/MS is one of the most commonly used methods in many laboratories for its accuracy and lower the limit of detection (LOD).

Though glyphosate can also be directly determined by LC–MS/MS in foods as the Europe reference method “Quick Polar Pesticides (QuPPE) method”, its sensitivity and selectivity are not satisfactory at the level 10 µg/kg for baby food [18,19]. As we all know, the fruits and vegetables are rich in sugar and pigments, and milk, cheese and butter to contain a lot of protein and fat. The complexity of these matrices greatly leads to interference in the target compound. To gain good sensitivity and selectivity, derivatization and purification by using solid phase extraction (SPE) have been used as current popular pretreatment methods. The most common approach to analysis of glyphosate and glufosinate always consists of derivatization with 9-fluorenylmethyl-chloroformate (FMOC), and the column of C18 was used for purification and concentration and subsequent detection with LC–MS/MS [20,21].

As mentioned, most methods are generally suitable for only one matrix or one kind of matrix such as water, plants, cereals and soil matrices [22–24]. It is being challenged to expand the scope of food samples and complex matrices and low concentration levels stand in the way. The ideal method needs to be suitable for a wide range of matrices, and easy to carry out for most laboratories, while being robust, selective, accurate and precise. In addition, the French Agency for Food, Environmental and Occupational Health & Safety (ANSES) launched the first French Infant Total Diet Study (iTDS), in order to estimate the dietary exposure and dietary risk to infants and young children below 3 years old in 2012. This program demanded to determine glyphosate and glufosinate residues in various food samples, which are including plant products with different water, protein and sugar contents, and animal product such as meat, cheese and milk. Although the maximum residue limits (MRLs) for two herbicides in food commodities have been established in European Union (10–20 µg/kg), the limit of quantification (LOQ) for baby food is required lower (5 µg/kg). Hence, it was imperious to develop a determination method for glyphosate and glufosinate applicable to a wide range food groups with high sensitivity and selectivity.

This study describes a laboratory validation of a LC–MS/MS method for the determination of low concentrations of glyphosate and glufosinate in various food products. The extraction with a solvent mixture including acidified water, methanol and dichloromethane, derivatization with 9-fluorenylmethylchloroformate (FMOC) and purification conditions by using solid phase extraction (SPE) column have been optimized. An isotopic method has been used for the quantification of two compounds by LC–MS/MS. The whole method has been validated and applied to 136 food samples provided by ANSES [25,26], such as milk-based baby foods but also “conventional” food products not specifically intended to infants and young children such as bread, biscuits, banana, apple, orange, potato, carrot, milk, cheese, butter, juice, and so on.

2. Material and methods

2.1. Chemicals and reagents

Glyphosate (purity 98%) and glufosinate (purity 97.5%) were provided by Cluzeau Info Labo (CIL, France) as certified analytical standards. Glyphosate (1,2-¹³C, ¹⁵N) and glufosinate (D3), used as internal standards, were also obtained from CIL. The solvents used were LC-grade acetonitrile (purchased from CARLO ERBA), formic acid, methanol (MeOH) and dichloromethane (from VWR). Other solvents and reagents were analytical grade such as disodium tetraborate decahydrate and ammonium acetate (NH₄Ac

(from SIGMA). SPE-C18 column was obtained from Waters and 9-fluorenylmethylchloroformate (FMOC) from SIGMA. Water was purified by the Milli-Q water purification system (Millipore).

Standard stock solutions of glyphosate and glufosinate (1 mg/mL) were prepared in the Milli-Q water and stocked in the plastic bottles at –20 °C. The intermediate solutions (100 µg/mL and 10 µg/mL) were prepared from standard stock solutions. The composite working solutions (1 µg/mL and 0.1 µg/mL) were diluted with Milli-Q water in the plastic bottles at 4 °C. FMOC solution was 20 mg/mL in acetonitrile. Disodium tetraborate was used for the borate buffer (0.05 mmol/mL). The 100 mL eluent solvent was prepared as following: 10 mL NH₄OH (20%), 40 mL the Milli-Q water and 50 mL methanol were added into a glass volumetric flask of 100 mL. A mixture of the two internal standards covered all the sample including calibration standard, matrix blank and samples at the concentration of 1 µg/mL.

2.2. Sample preparation

136 food samples came from ANSES. According to the European Union SANTE guideline [27], all the samples were divided into 3 groups for analysis (shown in Table 1). The samples with high water content (≥60%) accounted for 21% of the samples analyzed (mainly vegetables, fruits and drinks), and the ones with low water content such as biscuits or bread, for 13% of the samples. The products from animal origin, including milk-based baby foods represented 66% of the samples.

Some samples need to be chopped and homogenized in a mixer (model Retsch-GM200) according to the documentation “Commission Directive 2002/63/EC” [28]. All samples were stored at –20 °C until analysis.

2.3. Extraction procedure

An accurately weighed homogenized sample was put into a 50 mL plastic centrifuge tube. A mixture of solvents including Milli-Q water, acidified water, methanol and dichloromethane (shown in Table 2) were added into the tube. The tube was rotated mechanically for uniform mixing for 20 min and then centrifuged at 3000 rpm for 20 min at 4 °C or –15 °C (Rotanta 460R, Table 2). The aqueous extract of sample was transferred into a 20 mL plastic tube for derivatization. The DCM organic phase was removed.

For each analysis series, a calibration curve (5–10–25–50 µg/kg) in the same matrix group, a reagent blank, a matrix blank (free of analytes like glyphosate and glufosinate) and a spiked sample at 5 µg/kg were done.

2.4. Derivatization and purification procedure

The derivatization steps were performed as follows: 4 mL of the aqueous extract of sample was pipetted into a glass tube of 20 mL together with 50 µL of the labeled internal standard (1 µg/mL), 4 mL of the borate buffer (0.05 mmol/mL, pH 9) and 4 mL of FMOC reagent (20 mg/mL in acetonitrile) and the tube was swirled and left overnight at room temperature. After that, 3 drops of hydrochloric acid (HCl, 37%) was added into the tube and pH value was adjusted to 1.5.

A SPE-C18 cartridge (60 mg) was used to purify and concentrate the sample extract after derivatization. The cartridge was conditioned with methanol (3 mL) and then water solution (3 mL, with 0.1% formic acid). The total of the derivatized acidified extract was added into the cartridge and the cartridge was washed by acidified water (3 mL, with 0.1% formic acid) and dichloromethane (3 mL). The eluent (1 mL) was added and collected to a vial of 2 mL. The eluate was evaporated to dryness at 50 °C under the protection of nitrogen. The residue was dissolved with 200 µL of a solvent

Table 1

Classification, group and food product of analyzed samples.

Classification	Groups	Food products (sample number)
Plant products with high water content ($\geq 60\%$)	1. High water content (19 samples)	Bananas (1), kiwis (1), peeled pears (1), peeled apples (1), broccolis (1), carrots (1), zucchinis (1), cauliflowers (1), spinaches (1), beans (1), mixed vegetables (1), boiled potatoes (1*), fried potatoes (1), peas (1), leeks (1), pumpkins (1), mashed potatoes (1), vegetable soup (1), tomatoes (1)
	2. High acid content and highwater content (10 samples)	Oranges (1), clementines/mandarins (1), fruit juices (5), drinks with milk and fruits (1), diluted syrup extracted from fruits (1), sodas (1)
Plant products with low water content ($<60\%$)	3. High sugar and low water content (2 samples)	Fruit compote except apples (1), applesauce (1)
	4. High starch and/or protein content and low water and fat content (6 samples)	Fruit biscuits (1), natural or fruit cereals (1 ^a), brioche and brioche bread (1), chocolate bread (1*), chocolate mousse (1), chocolate cereals (1)
	5. "Difficult or unique commodities"(10 samples)	Cocoa powder (1), chocolate milk (1), chocolate spreads (1), Other products (7)
Products of animal origin	6. Meat (muscle) and seafood (1 sample)	Ravioli filled pasta (1)
	7. Milk-based baby foods and other milk products (80 samples)	Infant formulae for infants below 6 months (27), infant formulae (6–12 months) (35), follow-on formulae (9), other milk products (6), yogurt (3)
	8. Fat from food of animal origin (8 samples)	Cheese (5), cream (2), butter (1)

^a The residues of glyphosate or glufosinate were found in samples.**Table 2**

The extraction solvents for glyphosate and glufosinate.

Classification	Sample (g)	Acidified water (mL)	Methanol (mL)	Dichloromethane (mL)	Centrifugation temperature (°C)
High water content	5	20	–	20	4
Low water content	2.5	10	10	10	–15
Products of animal origin	5	10	10	10	–15

mixture (acetonitrile, the mobile phase A for LC–MS/MS and ammonium acetate at 0.05 mmol/L in water, the mobile phase B; 10/90; V/V). The final extract was then transferred into the conical vial for LC–MS/MS analysis after the ultrasonic shock.

2.5. Analysis by LC–MS/MS

2.5.1. Chromatographic conditions in liquid chromatography (LC)

Dionex UltiMate 3000 LC system was used and equipped with a C18 column (Atlantis 150 $\times$ 2.1 mm, 2.6 μ m, Waters). Acetonitrile was used as the mobile phase A and ammonium acetate at 0.05 mmol/L in water (pH = 5.0), as the mobile phase B. Correspondingly, the percentage of acetonitrile was changed linearly as follows: 0 min, 5%; 2.5 min, 5%; 8.0 min, 95%; 10.0 min, 95%, and 10.1 min, 5% with a total flow rate of 200 μ L/min and the injection volume of 5 μ L. The retention times for glyphosate-FMOC/glyphosate (1,2-¹³C,¹⁵N)-FMOC and glufosinate-FMOC/glufosinate (D3)-FMOC were respectively 7.0 and 7.2 min. The total run time was 20.0 min.

2.5.2. Mass spectrometry conditions

TSQ Quantiva mass spectrometer (Thermo Fisher Scientific) with an electrospray ionization (ESI) source was performed for MS analysis.

The mass spectrometer analyses were carried out in positive ion mode and the ion source type was heated electrospray ionization (HESI). The selective reaction monitoring (SRM) mode was used. The MS instrumental settings parameters were shown as below: spray voltage was 4000 V, the sheath gas and auxiliary gas pressures were 40 (Arbitrary) and 15 (Arbitrary) respectively, ion transfer tube temperature was 300 °C, and the vaporizer temperature was 250 °C. The capillary temperature was 320 °C and the collision gas (Ar) pressure was 1.5 mTorr. The obtained data were handled by Xcalibur software.

[M+H]⁺ ions (*m/z*) were selected as the precursor ions and two transitions were selected for each derivative (392.2 $\rightarrow$ 88.2 for quantification, 392.2 $\rightarrow$ 214.1 for qualification) of glyphosate-

FMOC and (404.1 $\rightarrow$ 136.1 for quantification, 404.1 $\rightarrow$ 208 for qualification) of glufosinate-FMOC. 395.2 $\rightarrow$ 91.1 transition of glyphosate (1,2-¹³C, ¹⁵N)-FMOC and 407.1 $\rightarrow$ 139.1 of glufosinate (D3)-FMOC were used for internal standard quantification.

3. Results and discussions

3.1. Pretreatment optimization

3.1.1. Extraction procedure

Based on the polarity and water solubility of glufosinate and glyphosate, water, acidified water (0.1% formic acid), methanol (MeOH), and dichloromethane (DCM) were used as extraction solvent in the matrix samples at spiked level 100 μ g/kg. When using just water as extraction solvent, the recoveries were not high, even only 50% for glyphosate in the milk as Fig. 1 shown. The choice of an acidified water (0.1% formic acid) as extraction solvent was dictated by the very polar character of both glyphosate and glufosinate and their insolubility in most organic solvents. Although the recovery can be improved with acidified water (0.1% formic acid), the recovery was not satisfied with milk sample and interferences persisted and affected the analyst. The main problem with this procedure is the presence of other water-soluble components in the extract that will hamper the analyte. Therefore, two other solvents were added to remove interferences: methanol for the removal of protein and starch precipitation (only for the matrix in medium and high protein content such as cereals, meat, . . .) and dichloromethane (DCM) (for all the matrix) for the removal of hydrophobic matrix components or non-ionic low to medium water solubility matrix components such as lipids. A simultaneous process of extraction and purification was used to minimize the sample handling time. Adding dichloromethane and methanol for extraction improved the recovery of glyphosate from 50% to 92% and glufosinate from 70% to 98%. After a series of experiments, the proportion between acidified water (0.1% formic acid), methanol and dichloromethane were chosen which is shown in Table 2.

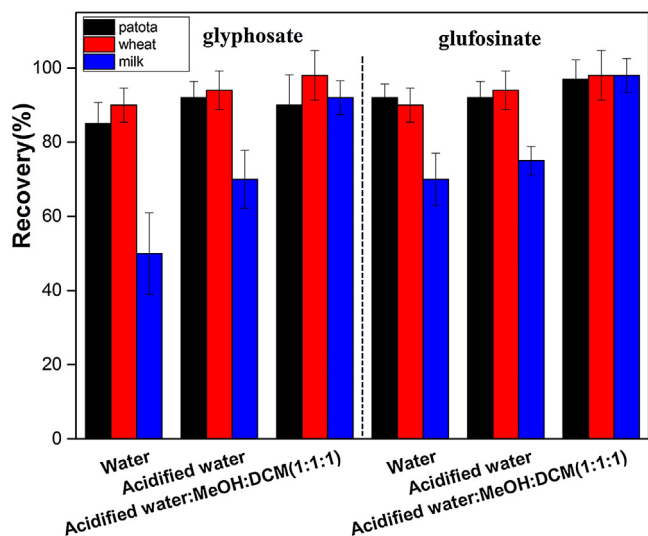


Fig. 1. Influence of extraction solvent on the recoveries for glyphosate and glufosinate.

3.1.2. Derivatization conditions

Because of high polarity, retention in reversed phase columns and sensitivity in ESI/MS is either low in the case of underivatized glyphosate and glufosinate. Since the work of Moye and Boning [30], who were the first to use FMOc for glyphosate derivation, many procedures have been reported in the literatures [20,21]. After derivatization, analytes' polarity is significantly reduced enabling better retention on the reversed phase and thereby better separation from matrix components. In addition, the ionization efficiency of analytes in ESI source was enhanced by its hydrophobic parts in molecules and limit of detection (LOD) was improved. Therefore, the FMOc was used as derivatizing reagent in this method. In terms of reaction time, data ranged from 5 min [21] to overnight [24]. The effect of different derivatization reaction times and temperatures was studied. With the increase of reaction time, the recovery rate increased significantly. The maximum reaction yield was observed after 4 h (shown in Fig. 2). In practice, 12 h was chosen to be reaction time. An influence of the reaction temperature was observed as Fig. 2b. Higher temperature reaction tended to decrease the recovery of product. It is the same as reported by Druart [11]. Therefore, the derivatization was completed at room temperature.

The effect of FMOc concentration was not investigated in our study but was fixed at 20 mg/mL. The use of FMOc in excess was to compensate for high reactivity with water leading to the formation of FMOc-OH products and matrix components-primary and secondary amines.

3.1.3. SPE purification

To remove the interferences, excessive FMOc and concentrate the analytes to increase sensitivity, a SPE-C18 cartridge was chosen for purification and concentration. Firstly, methanol and acidified water were used to condition the cartridge. Once the derivatization reaction took place overnight, hydrochloric acid was added to stop the reaction, by low-ering the pH. Then the derivatized acidified extract was added into the C18 cartridge. Acidified water and DCM were chosen for rinsing to remove the excess FMOc and some polar interference. In order to carefully check for possible analyte losses during dichloromethane washing, the solution was also analyzed from washing step with water and dichloromethane and the result revealed no indication of analyte losses. The eluent at pH 9 was selected to elute the analytes from the cartridge. The eluate was then evaporated to dryness and the residue was dissolved with a

solvent mixture of the mobile phases of LC-MS/MS to reduce solvent effect. Finally, the sample extract was concentrated 20 times (from 4 mL to 0.2 mL) before the analysis on LC-MS/MS. Following the above steps, the recovery of glyphosate could reach 92% and one of glufosinate, 94% in the milk.

A combination of several steps as indicated below makes our method more sensitive and selective: an efficient extraction with a solvent mixture, the FMOc derivatization, a SPE-C18 cartridge for purification and the concentration 20 times (from 4 mL to 0.2 mL) for sample extract before instrument analysis and finally, the determination by LC-MS/MS with its high sensibility and selectivity. Therefore, the combination of all these factors leads to the high sensitivity and selectivity of our method.

In addition, a calibration curve in the same matrix group was used to quantify glyphosate and glufosinate in the presence of the labeled internal standard (glyphosate ^{13}C , ^{15}N and glufosinate D3) to reduce the matrix effect.

3.2. Method validation

Validation of the developed method was performed as described in the following documents: European Union SANTE guideline [27] and French Standard NF V03-110 [29]. The calibration curves were obtained by the linear analysis ($Y = aX + b$ and linear $1/X$) at matrix-matched calibration, which was conducted at four levels ranging between 5, 10, 20 and 100 $\mu\text{g}/\text{kg}$ with two replicates for five days. The accuracy was evaluated by the recoveries (%) and the precision was expressed by RSD_R (%) of the spiked samples. The application fields, the limit of detection (LOD), the limit of quantification (LOQ), and the accuracy and precision of the method were discussed respectively as below.

3.2.1. Choice of the food groups and food categories

Glyphosate and glufosinate herbicides are used at pre- and post-emergence weed control, and their residues are theoretically mainly expected in cereals, which were divided into the commodity groups with high starch and/or protein content and low water and fat content shown in Annex A of SANTE guideline [27]. Consequently, maize, rice, wheat and barley were selected as matrix blank to validate the developed method. Considering the diversity of food products to be tested, the verification of method in different matrices spiking samples was also done, as shown in method application. The linearity, the limit of detection (LOD), the limit of quantification (LOQ), and the accuracy and precision of the method were discussed respectively as below.

3.2.2. Linearity, LOQs and LODs

The linear range was determined by two replicate analyses of calibration standards (matrix-matched calibration), which was conducted at four levels ranging between 5, 10, 20 and 100 $\mu\text{g}/\text{kg}$ with internal standard mixture for five days. The calibration curves were obtained by the linear analysis ($Y = aX + b$ with $1/X$). In five matrices (2 maize samples, 1 rice, 1 wheat and 1 barley) the coefficient of determination (R^2) for the standard curves of glyphosate and glufosinate were greater than 0.99 as attachment material A1.

In theory, the LODs and LOQs are obtained by the ratio of the signal-to-noise (S/N) of 3 and 10 from the lowest concentration levels of the spiked samples. However, the results of reproducibility and repeatability were not satisfying in practical application using the value of 10 S/N as LOQ. In this study, although the target LOQ for glyphosate and glufosinate was 10 $\mu\text{g}/\text{kg}$, the LOQs for the two pesticides were chosen at 5 $\mu\text{g}/\text{kg}$ for French Infant Total Diet Study. The experiment was carried out for duplicate in the five matrices during five days. We used three parameters (accuracy, fidelity and

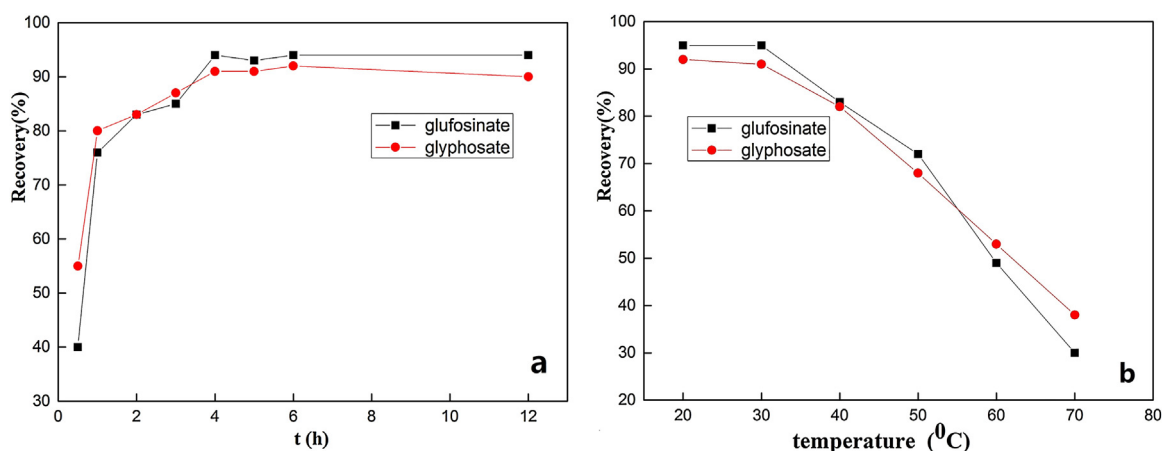


Fig. 2. Influence of derivatization time and temperature on the recoveries for glyphosate and glufosinate.

Table 3

Accuracy and precision of the proposed method in studied matrices.

Matrix, day	[C ^a]	Recoveries (%)										Avg.R(%)	RSD _R (%)
		Maize, day1		Maize, day2		Rice, day3		Wheat, day4		Barley, day5			
glyphosate	5	100	94	106	100	114	105	109	100	108	103	104	5.60
	10	97	102	109	107	99	98	112	109	101	107	104	6.38
	20	99	103	100	95	101	108	100	93	106	99	100	4.53
Glufosinate	5	100	98	107	108	109	106	106	103	96	92	103	5.88
	10	107	98	95	94	99	103	108	108	102	105	102	5.12
	20	91	96	101	103	98	99	98	94	100	100	98	3.82

^a Which is spiked level (μg/kg).

homogeneity) to test LOQ. The accuracy, fidelity and homogeneity were calculated as following:

- (1) Accuracy criterion ($\leq 10\%$): $\frac{|C_s - C_M|}{SD \times \sqrt{10}}$, where C_s was the chosen value; C_M , mean value and SD , standard deviation
- (2) Fidelity criterion ($\leq 20\%$, relative standard deviation, RSD): $SD/C_M \times 100\%$
- (3) Homogeneity: the homogeneity of data was tested by SHAPIRO-WILK statistic.

The values obtained of accuracy were 2.2% and 1.3% for glyphosate and glufosinate respectively. Two values obtained of fidelity for glyphosate and glufosinate were both 5.7% and they reached the requirements of the precision. According to the test of SHAPIRO-WILK, the calculated results demonstrated the data followed a normal distribution. Consequently, it was satisfying with the homogeneity of data. Thus, the chosen LOQs of glyphosate and glufosinate at 5 μg/kg were suitable in the selected matrices. The LODs of glyphosate and glufosinate were 1.7 μg/kg correspondingly ($LOD = 3/10 \text{ LOQ}$). In addition, it is observed that the values of the ratio S/N for glyphosate and glufosinate were higher than 1900 and 20000 respectively at 5 μg/kg (far more than 10 S/N), which shows that in practice a lower LOQ can be achieved for glufosinate. The results were shown in attachment material A2.

3.2.3. Accuracy and precision

In the validation plan, we used an accuracy profile to validate a method. The details as follows:

The number (n) of series, repetition and levels for the validation plan were chosen (repeatability: $n = 2$, reproducibility: $n = 5$, level: $n = 3$, level of concentration: LOQ, 2 LOQ and 4 LOQ) according to the guidance document SANTE (repeatability: $n \geq 1$, reproducibility: $n \geq 5$, level: $n \geq 2$, level of concentration: LOQ and 2 LOQ or 10

LOQ) and NF V03-110 (repeatability: $n \geq 2$, reproducibility: $n \geq 3$, level: $n \geq 3$, level of concentration: no request).

First, the accuracy was evaluated by absolute mean bias, relative mean bias (%) and average recovery (%). The precision was expressed by standard deviations (SD) of repeatability, inter-series & intermediate fidelity and relative standard deviation (RSD_R %). The recoveries for glyphosate ranged from 93% to 114% in inter-day studies and the average recoveries (%) for three levels (5, 10 and 20 μg/kg) were 104, 104 and 100 respectively. Accordingly, the recoveries for glufosinate ranged from 91% to 109% in inter-day studies and the average recoveries (%) for three levels (5, 10 and 20 μg/kg) were 103, 102 and 98 respectively in Table 3. The RSD_R % values for glyphosate and glufosinate ranged from 3.8 to 6.1 in inter-day, which reached the evaluation requirements for the precision. The uncertainty (1) included relative mean bias (RMB) for accuracy and RSD for precision ($1 = \sqrt{(RMB)^2 + RSD^2}$), and it ranged from 9% to 16% for glyphosate and 9% to 13% for glufosinate.

Second, a tolerance interval of expectation is constructed which contains a probability β % (80% chosen). To use an accuracy profile to validate a method, the two decision criteria (limits of acceptability and probability β) was established as follows. The result was shown in attachment material A3 and attachment material A4.

- (1) Limits of acceptability. They serve to translate the practical objectives of the users and delimit an interval around the reference value. In most cases, these limits are regulated or derived from regulation. In this study, the recovery (between 70 and 120%) for accuracy and RSD_R limits ($\leq 20\%$) were used according to the guidance document SANTE.
- (2) Probability β . It represents the proportion of future average results within the tolerance intervals. The value chosen for β depends largely on the application field. It is obvious that the smaller β , eg. 70%, the more likely the method is to produce

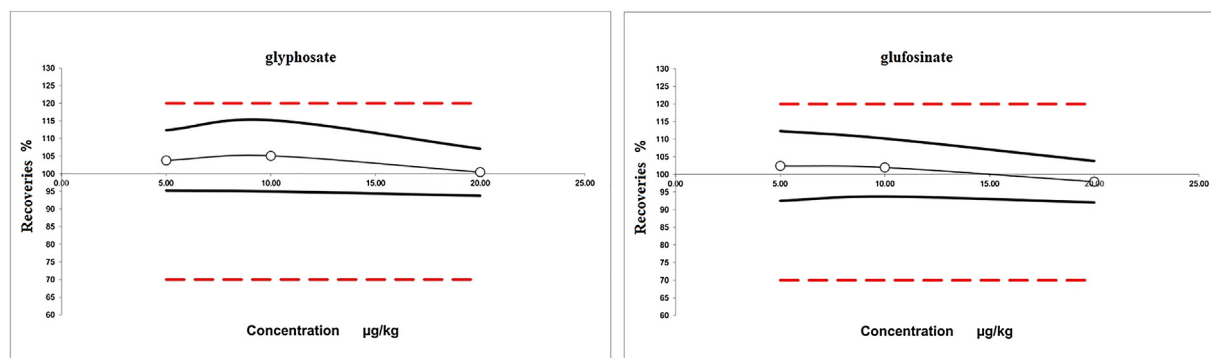


Fig. 3. Accuracy profile of glyphosate and glufosinate for the proposed method.

Table 4

The recoveries of glyphosate and glufosinate in expanded matrices.

Matrices	[C ^a]	Recoveries (%)										Avg.R(%)	RSD _R (%)
		Banana		Potato		Milk		Cheese		Yogurt			
glyphosate	5	94	106	102	94	96	96	108	90	96	82	97	7.9
	10	103	96	98	103	97	88	100	105	102	89	98	6.1
Glufosinate	5	104	100	90	112	96	108	104	97	106	102	104	7.3
	10	97	99	99	95	83	88	116	92	96	101	95	6.0

^a Which is spiked level (µg/kg).

results that do not correspond to the announced specifications. In NF V03-110 this probability β has been set at 80%, at least.

Based on the accuracy, precision, uncertainty and a tolerance interval of expectation, the accuracy profile of the method was calculated and the results were shown in Fig. 3. All different combinations complied with the acceptable method performance parameters suggested by SANTE guide line. The experiment results indicated that the developed method was in the satisfying range expected for analysis.

3.3. Application

3.3.1. Application fields

This developed method has been certified by the French certification organization (COFRAC), and has been applied to the analysis of 136 food samples provided by ANSES in January 2013, and currently for routine national monitoring. The milk-based baby foods were mainly provided “as consumed” (liquid) by ANSES and analyzed as such by the laboratory. The food samples were divided into three groups as shown in Table 1.

3.3.2. Verification in expanded matrices

Considering the variety of food products to be tested (Table 1), the verification of the method in expanded new matrices spiking samples was also conducted according to SANTE guideline. The plant products with low water content (<60%) have been validated. So according to three kinds of actual samples, the different matrices were selected for verification including fruit (banana) and vegetable (potato), milk, yogurt and cheese. The average recoveries (%) and RSD_R (%) were calculated by the spiked samples including five kinds of matrix in two replicates at two levels (5 and 10 µg/kg). The recoveries of glyphosate and glufosinate ranged from 82% to 116% (between 70 and 120%) and the RSD_R, from 6.0 to 7.9% ($\leq 20\%$). The result was displayed in Table 4. So, the method was verified in these application fields with acceptable accuracy and precision. Meanwhile, the chromatograms at the level (5 µg/kg) in 5 different spiked matrices (banana, potato, cheese, milk and yogurt) and a blank matrix (free of analytes like glyphosate and glufosinate) were

shown in Fig. 4. The 4 peaks are respectively shown in the chromatogram for each matrix: the quantitative peak of glyphosate (top left) and one of its internal standard (bottom left), and one of glufosinate (top right) and one of its internal standard (bottom right). The last chromatogram (bottom right) was the blank matrix. From the figure, it displayed little interference in the matrices. Therefore, it indicated that the two pesticides in these different matrices can be determined, and that the developed method was sensitive enough to quantify glyphosate and glufosinate in foods.

3.3.3. Analytical results

To ensure the quality of the analytical results, a reagent blank, a matrix blank and two spiked samples at 5 µg/kg (QC) for two compounds and at 2 µg/kg for glufosinate (as requested by ANSES) were analyzed in each kind of matrix and each series experiment, for the 136 food samples analyzed by this method. Finally, among these samples, only 3 samples contained quantified levels of glyphosate or glufosinate residues. Glyphosate residues were quantified at 6.0 and 34 µg/kg in two breakfast cereal samples respectively. Glufosinate residues were quantified at 9.8 µg/kg in a sample of boiled potatoes. No residues were detected in the other samples including milk-based baby foods, with LOD ranging from 1 µg/kg for glufosinate to 2 µg/kg for glyphosate as requested by ANSES. For these samples, in practice, better analytical limits were verified for each analytical batch and the results were satisfied. In order to better monitor the dietary exposure of infants and young children to these two compounds and their related residues (e.g. trimethyl-sulfonium, AMPA, acetyl-glyphosate and acetyl-AMPA), further analytical developments and most frequent analysis should be undertaken in the framework of monitoring programs, especially for cereals and potato products.

4. Conclusion

A reliable and sensitive method was developed to detect and quantify glyphosate and glufosinate residues in food by LC–MS/MS. High accuracy and precision could be achieved by off-line derivatization and purification. Good validation parameters for linearity, recovery, LOD and LOQ were acquired and the verification results

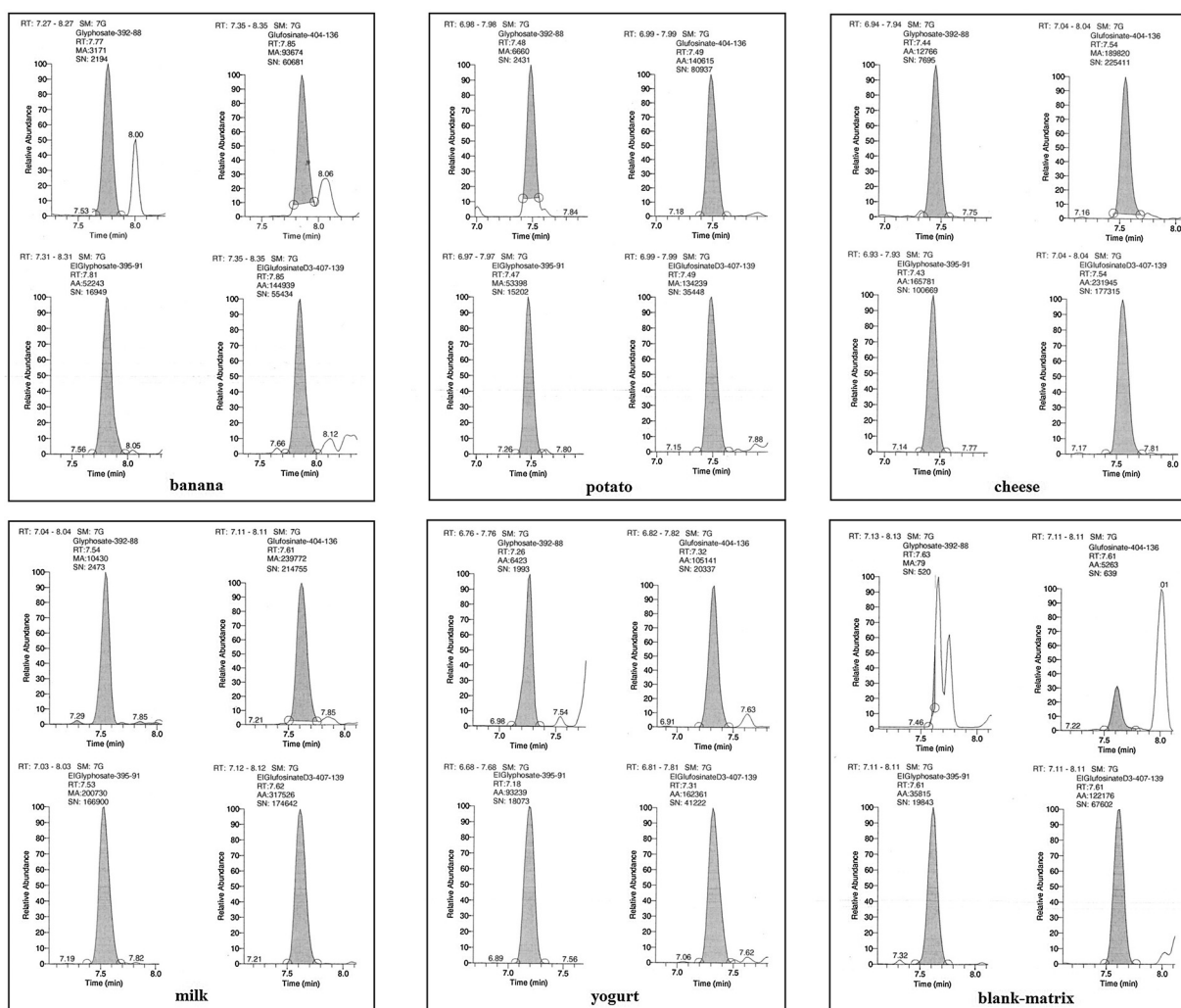


Fig. 4. Chromatograms (LC-MS/MS) in expanded matrices at the level (5 µg/kg).

demonstrated extensive applicability of the method for the analysis of glyphosate and glufosinate in various food products. The method can be used not only for high water content samples, but also for samples with low one, high protein and products of animal origin. In conclusion, the developed method can be a dependable and sensitive one for the routine monitoring of glyphosate and glufosinate in food. Analytical perspectives should include the development of analytical methods to better monitor glyphosate related metabolites in food, e.g. trimethyl-sulfonium, AMPA, acetyl-glyphosate and acetyl-AMPA.

Conflict of interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.chroma.2018.03.036>.

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